- **Primary Hyperparathyroiodism:** Parathyroid hormones increases the synthesis of 1,25, dihydroxy vitamin D3 (calcitrol), which increases calcium absorption in the intestines and reabsorption from the kidney tubules.
- **Recumbency:** Prolonged bed rest, particularly when a patient is immobilized, leads to generalized decalcification of the bones. The breakdown products of bones are excreted through the urine, and may form urinary stones if liquid intake is scanty.
- **Congenital malformations and infections:** Congenital malformations of the renal pelvis, and repeated urinary tract infections predispose to kidney stones formation.

10.4.2. Types of kidney stones

The most common constituents of kidney stones are oxalates, urates or phosphates, combined with calcium.

Calcium: The prevalence of calcium stones is higher in those taking a low fibre diet. A vegetarian diet predisposes to urinary bladder stones while meat diet is claimed to predispose to kidney stones. Increasing the protein intake enhances urinary calcium excretion in normal subjects as well as in those with urinary stones.

Oxalate: Crystals of calcium oxalate are found mostly in acid urine, but may also be present in neutral or oxalate urine; they may not be associated with increased urinary calcium excretion. Ingestion of rhubarb, spinach, tomatoes, strawberries, chocolate or tea produces temporary increase in oxalates in all individuals (secondary hyperoxaluria). It is possible to decrease the oxalate content of vegetables by using phosphate fertilizer.

Uric acid: Uric acid stones are common in patients with gout and are also seen in the tropics.

Phosphate: Phosphate stones tend to form in the bladder when there is infection. Some infecting organisms split urea into ammonia, and phosphates precipitate in the strongly

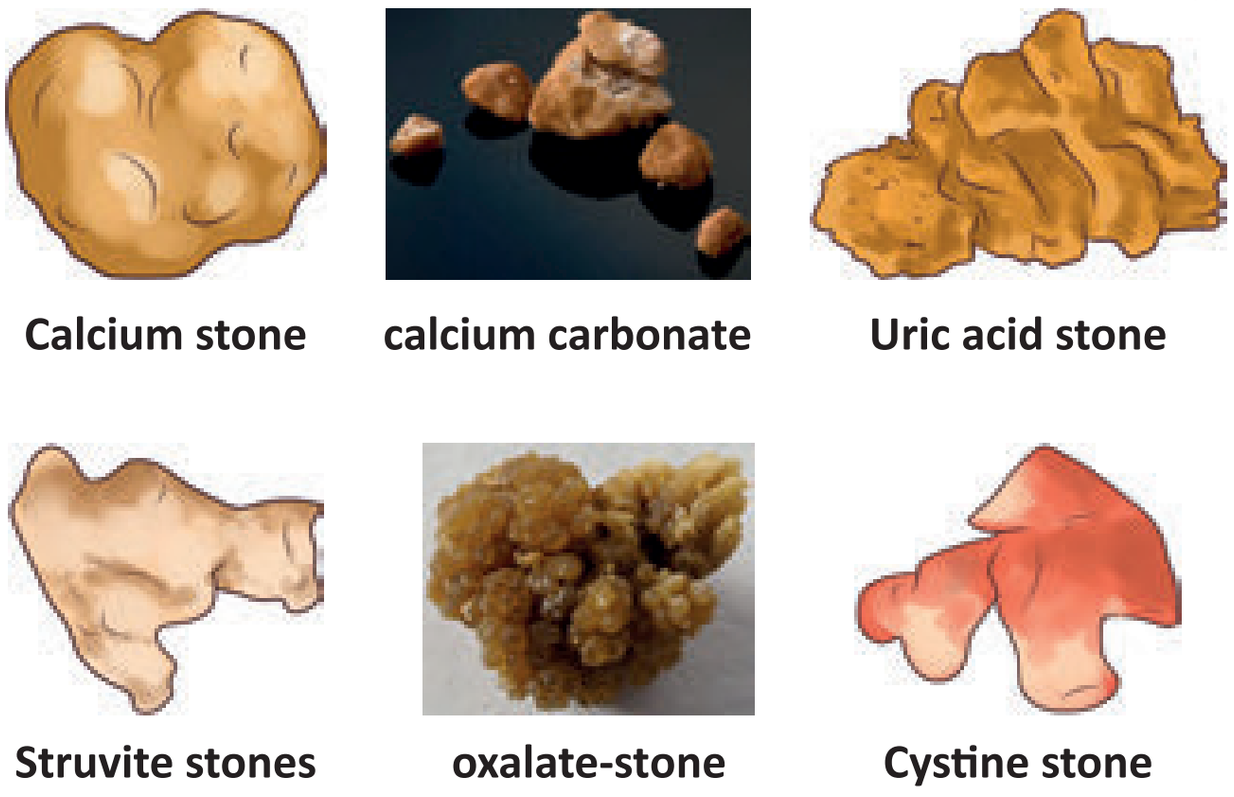


Fig 10.5 Types of kidney stones

Unit 10 Diet in kidney diseases Final 12-12-2018.indd 166 25-02-2019 15:37:57